(ST 541) Final Project

Introduction

Most fields of research struggle to differentiate between exploratory and confirmatory analyses, or they recognize the distinction but fail to respect the implications of the two. This project addresses the critical issue of conflating exploratory analysis with confirmatory analysis, which involves hypothesis testing and statistical inference. Tong (2019) highlights that routines often employed during exploratory analyses that can lead to overfitting the data, resulting in biased final models then subsequent statistical inference are often made that underestimated uncertainties and provide misleading statistical significance results.

Exploratory analyses typically involves model selection, where the goal is to identify the model that best explains the underlying data-generating process. While exploratory analyses can take various forms, this project focuses specifically on model selection within the framework of regression. Several aspects of this issue are explored, including:

1) A reproduction of the well-documented phenomenon where model selection introduces biased sampling distributions of estimates, resulting in a mixture distribution.

2) A showcase of how coefficients close to zero are often treated as exactly zero during model selection, even when a genuine relationship may exist, and the implications of this.

3) A look at how regularization techniques (LASSO and Ridge) affect the sampling distributions.

4) Finally, a demonstration of a potential solution to these issues via data splitting.

Through these explorations, this project seeks to highlight the practical consequences of conflating exploratory and confirmatory analyses in these settings, as well as to propose strategies for improving the reliability of statistical inference in model selection contexts.

Simulations

The simulations in this project aim to evaluate the impact of various model selection techniques on the sampling distribution of various regression statistics. The provided code (available in the Appendix) simulates a regression framework where model selection techniques such as forward, backwards, stepwise selection, LASSO, and Ridge can be applied to a dataset. Through these simulations, the highlighting of how these methods affect the accuracy and distribution of the estimated coefficients is shown. The function simulates multiple datasets under the true model and evaluates the resulting parameter estimates through visualization of their distributions using density plots or histograms.

What follows is the exploration of the various aspects described in the introduction under differing parameter settings to display potential differences. All of the following are done with 10,000 simulations and forward selection criterion (not including regularization techniques) uses AIC, all other function arguments may vary. All mean vector elements are unchanged throughout due to no effect on coefficients (aside from the intercept).

Figures for simulations, in order of appearance, can be found in the Appendix.

1) A replication of the well-known phenomenon where model selection leads to a estimate mixture sampling distributions.

- **Parameter Setting 1:** The explanatory variables $\boldsymbol{X}$ follow a multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, as shown below:

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \begin{pmatrix}1\\1\\1\end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix}1.00 & 0.50 & 0.50\\0.50 & 1.00 & 0.50\\0.50 & 0.50 & 1.00\end{pmatrix}\right), \quad \boldsymbol{\beta} = \begin{pmatrix}2\\-0.5\\0\\0.5\end{pmatrix}$$

$$\boldsymbol{Y} = \boldsymbol{X\beta}, \quad n = 50, \quad \boldsymbol{\epsilon} \sim N(0, 16)$$

- **Parameter Setting 2:** The same structure as Parameter Setting 1, but with a larger sample size of $n = 300$.
- **Parameter Setting 3:** Again, the same structure as Parameter Setting 1, but now the covariance structure of $\boldsymbol{X}$ is modified such that the explanatory variables (X_i) are uncorrelated:

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \begin{pmatrix}1\\1\\1\end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}\right)$$

- **Parameter Setting 4:** Again, the same structure as Parameter Setting 1, but now the error terms are more variable

$$\boldsymbol{\epsilon} \sim N(0, 36)$$

2) An examination of how coefficients near zero are often treated as exactly zero during model selection, even when a true relationship may exist.

- **Parameter Setting 1:**

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \mathbf{1}_5, \boldsymbol{\Sigma}_{ij} = \begin{cases}1 & \text{if } i = j,\\0.5 & \text{if } i \neq j.\end{cases}\right), \quad i, j = 1, \ldots 5$$

$$\boldsymbol{\beta}^T = \begin{pmatrix}2 & 0.0625 & 0.125 & 0.25 & 0.5 & 1\end{pmatrix}, \quad \boldsymbol{Y} = \boldsymbol{X\beta}, \quad n = 50, \quad \boldsymbol{\epsilon} \sim N(0, 16)$$

- **Parameter Setting 2:** The same structure as Parameter Setting 1, but with a different set of coefficients demonstrating similar results

$$\boldsymbol{\beta}^T = \begin{pmatrix}2 & -0.25 & -0.125 & 0 & -0.125 & 0.25\end{pmatrix}$$

3) An display of how regularization methods, such as LASSO and Ridge, influence the sampling distributions of estimates.

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \mathbf{1}_6, \boldsymbol{\Sigma}_{ij} = \begin{cases}1 & \text{if } i = j,\\0.5 & \text{if } i \neq j.\end{cases}\right), \quad i, j = 1, \ldots 6$$

$$\boldsymbol{\beta}^T = \begin{pmatrix}2.0 & 0.125 & 0.25 & 0.50 & 1.0 & 2.0 & 3.0\end{pmatrix}, \quad \boldsymbol{Y} = \boldsymbol{X\beta}, \quad n = 50, \quad \boldsymbol{\epsilon} \sim N(0, 25)$$

4) A demonstration of a potential solution to these challenges through the use of data splitting.

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \begin{pmatrix}1\\1\\1\end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix}1.00 & 0.50 & 0.50\\0.50 & 1.00 & 0.50\\0.50 & 0.50 & 1.00\end{pmatrix}\right), \quad \boldsymbol{\beta} = \begin{pmatrix}1\\-3\\-1\\-2\end{pmatrix}$$

$$\boldsymbol{Y} = \boldsymbol{X\beta}, \quad n = 50, \quad \boldsymbol{\epsilon} \sim N(0, 25)$$

Results

1) For each simulation the number of possibly chosen models is $2^{\text{number of covariates}}$ (not considering interactions for simplicity). For the initial set of simulations under Parameter Setting 1, covariates were labeled alphabetically ($X_1 = A, X_2 = B, X_3 = C$), The model selection frequencies are summarized in the following table.

Model	Chosen %	Chosen Freq.	$\hat{\beta}_0$	$\hat{\beta}_0^*$ (Sel.)	$\hat{\beta}_1$	$\hat{\beta}_1^*$ (Sel.)	$\hat{\beta}_2$	$\hat{\beta}_2^*$ (Sel.)	$\hat{\beta}_3$	$\hat{\beta}_3^*$ (Sel.)
`Null`	55.77%	5577	2.00	2.00						
`A`	11.31%	1131	2.25	2.78	-0.25	-0.79				
`AB`	3.12%	312	2.17	2.13	-0.34	-1.20	0.17	1.05		
`ABC`	1.16%	116	2.00	1.93	-0.50	-1.45	0.01	0.01	0.49	1.50
`AC`	7.05%	705	2.01	1.93	-0.50	-1.44			0.50	1.51
`B`	8.16%	816	2.00	1.98			0.00	0.03		
`BC`	2.77%	277	1.84	1.83			-0.16	-0.95	0.32	1.14
`C`	10.66%	1066	1.76	1.24					0.24	0.79

Table 1: Model Selection Breakdown with Coefficients

What Figure 1 displays is the combined sampling distribution of all the different estimates for all possible models above. For example, in Figure 1 the top left subplot represents the combined sampling distribution across all models for $\hat{\beta}_1$ (in purple) and $\hat{\beta}_1^*$ (in blue). The distinction between the two is warranted due to the coefficients estimating different things, in fact within each column a different parameter is being estimates due to the presence or absence of others. We see that since this model combination occurs the result is a mixture sampling distribution. Notably, the model selection sampling distribution almost never identifies the true value of the beta coefficient (illustrated with a dashed vertical line).

Figure 2 separates the samplings distribution shown in Figure 1 for β_1^* for all models that include the coefficient. We see that there is still a mixture distribution in the incorrect models and even the correct model, `Model AC`, is biased from the generated β_1 value. This is from the model selection process as well since the models that have been selected here are not a representative sample of all of the true, say possible `Model AC`s. This point is displayed more in Table 1 where the average coefficient values for no model selection models are displayed (in purple) compared to (blue) the average coefficients from performing model selection. Two main points of interest here, one is that even without model selection we see that the incorrectly specified models have bias coefficients, this is due to the fact that correct correlated covariates are missing. Lastly and perhaps more importantly we see that the all of

the post-model selection coefficients are biased, their magnitudes are often much greater due to the fact that lower magnitude coefficients are excluded from the sampling pool from model selection - if the coefficient is near zero the covariate is likely to be dropped from the model.

Figures 3 and 4 show similar results and Figure 3, since $\beta_2 = 0$ meaning it should not be included in the model, we see some very high levels of bi-modality centered around 0. This breakdown between models holds for the various other settings explored but is not expanded upon again due to length restraints and for fear of much repetition.

Circling back to Figure 1, both the coefficient estimates and the corresponding t-statistics are displayed. While either metric provides a useful summary of the distortion, the t-statistic has the added advantage of incorporating the standard error of the coefficient.

Continuing on to the other parameter settings, when we increase the sample size by a factor of six, the bi-modality of the sampling distribution diminishes, as shown in Figure 5. However, a bias remains clearly evident, demonstrating that the issue is not completely resolved even with a larger sample size.

Turning to Parameter Setting 3, where the explanatory variables are uncorrelated, Figure 6 demonstrates that the problem persists. The bi-modal nature of the mixture distribution is still apparent, highlighting that lack of correlation between predictors does not mitigate this issue. In fact, in relation to the "correct" sampling distribution it is much worse than before because the "correct" sampling distribution is more clustered around the true parameter.

Finally, under Parameter Setting 4, Figure 7 illustrates the impact of increased error variance. In this case, the bi-modal nature of the mixture distribution becomes even more pronounced. This suggests that higher variability in errors exacerbates the challenges associated with identifying the true beta coefficient in this simulation framework.

2) For parameter settings 1, Figure 8 illustrates that as the true coefficient value decreases and approaches zero, the bimodality of the model selection sampling distribution becomes more pronounced.

   In Figure 9, we modify the parameter settings such that the true coefficients are centered around and include zero. The results confirm a similar pattern: as the true coefficient approaches zero, the bimodality intensifies, peaking when the coefficient is exactly zero. This outcome highlights the largest discrepancy in coefficient estimates, stemming from the model's tendency to either include or exclude the covariate entirely.

3) The inclusion of LASSO and Ridge regression, shown in Figure 10, highlights the effects of regularization on the sampling distributions of estimated coefficients.

   LASSO produces a biased sampling distribution, especially for coefficients close to zero, as its penalty can shrink these coefficients to exactly zero. This behavior, while useful for variable selection, leads to a systematic distortion of the sampling distribution and an overrepresentation of close-to-zero estimates. This effect becomes less prominent when the coefficient values grow to a certain point.

   Ridge regression, on the other hand, does not perform variable selection but still biases the coefficients. Its penalty reduces all estimates proportionally, resulting in a more evenly biased sampling distribution compared to LASSO. One might notice that the density plot of $\hat{\beta}_4$ of Ridge is closely centered on the true value, I believe this to be competing biases that happen to cancel out in this scenario supported by the fact that all the other plots show a bias.

4) Figures 11 and 12 are of various data splitting allocations, 50-50 split training and test and then a 75-25 split respectively. At first these results seem to suggest that the advice of splitting the data does not work, but remember that what is shown in these figures is a combined sampling distribution over many possible models as such there are many different "true" parameters being tested and combined into each density plot.

   One can, like before, separate these mixtures, since we are in a simulation setting, and look at the sampling distribution of coefficient estimates for each model individually. Figures 13, 14, 15 do just this, each for a coefficient. We see the purple and green densities represent if no model selection were to occur on the training set and test set respectively and they are almost identical which we would expect since they are estimating the same thing. Then we turn our attention to the blue density which represents the training data used for model selection and fitted with the chosen model, as before see a bias and sometimes bi-modality present. Lastly the red density represents if we were to use the training data for model selection and fit the chosen model using the test and we can see that it matches the densities of the datasets that did not use model selection and as such is an unbiased sampling distribution of the coefficient of interest for that model.

   Figures 16, 17, 18 are the decomposition of Figure 12 with the 75-25 data split and we see similar results except now since the test set has a smaller sample size we see the precision is lower, but it is still unbiased as before.

   Thus we see that through the use of this data-splitting, uses the training data to perform model selection and fitting the chosen model using the testing data, we can obtain a representative representation of our coefficient sampling distribution and avoid the post-model selection bias that is present when using the same data to perform model selection and the model fitting.

Conclusion

This project underscores the critical importance of distinguishing between exploratory and confirmatory analyses in statistical research, particularly within the context of model selection. The simulations and analyses conducted herein provide compelling evidence of the adverse effects of conflating these two types of analyses, focused on the biased sampling distributions which leads to incorrect statistical inference.

The exploration of different parameter settings revealed how model selection introduces biased mixture distributions of estimates highlighting the challenges in statistical inference post-model selection. This project highlights the necessity of completely prespecifying the study design and statistical analysis plan prior to engagement with the data. By adopting practices that respect the distinctions between exploratory and confirmatory analyses, researchers can enhance the robustness and credibility of their findings, ultimately leading to better science.

References

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Appendix

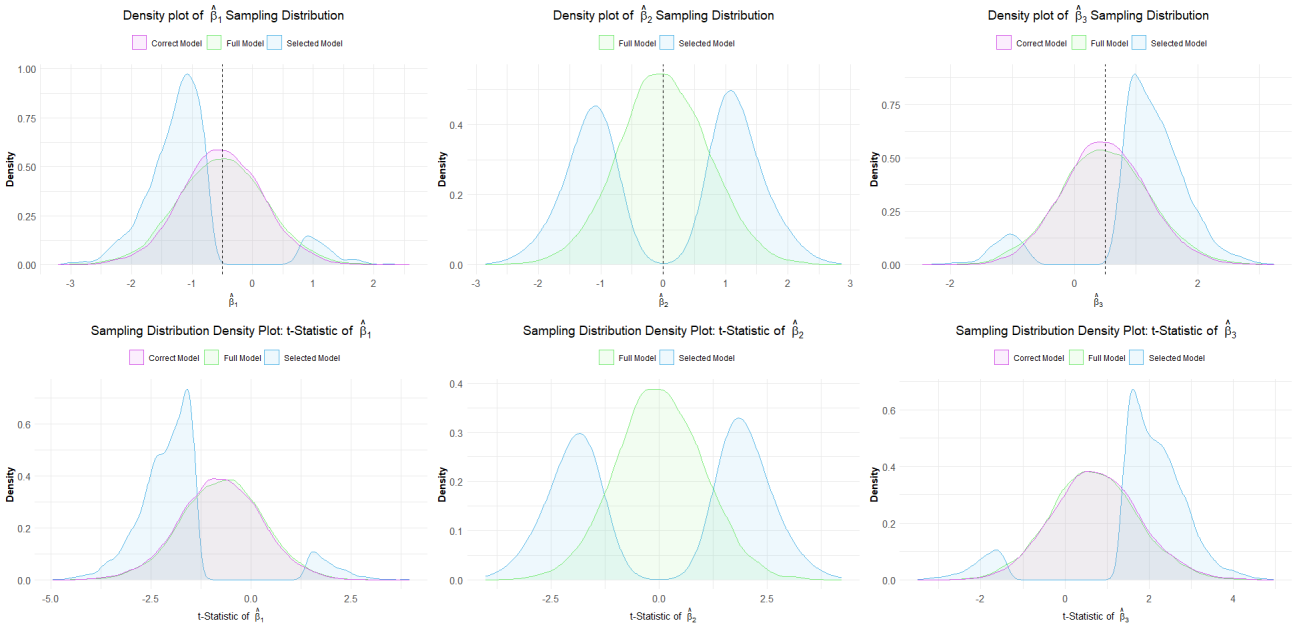

Figure 1: Parameter Setting 1 - Mixture of each coefficient and t-statistics across chosen models

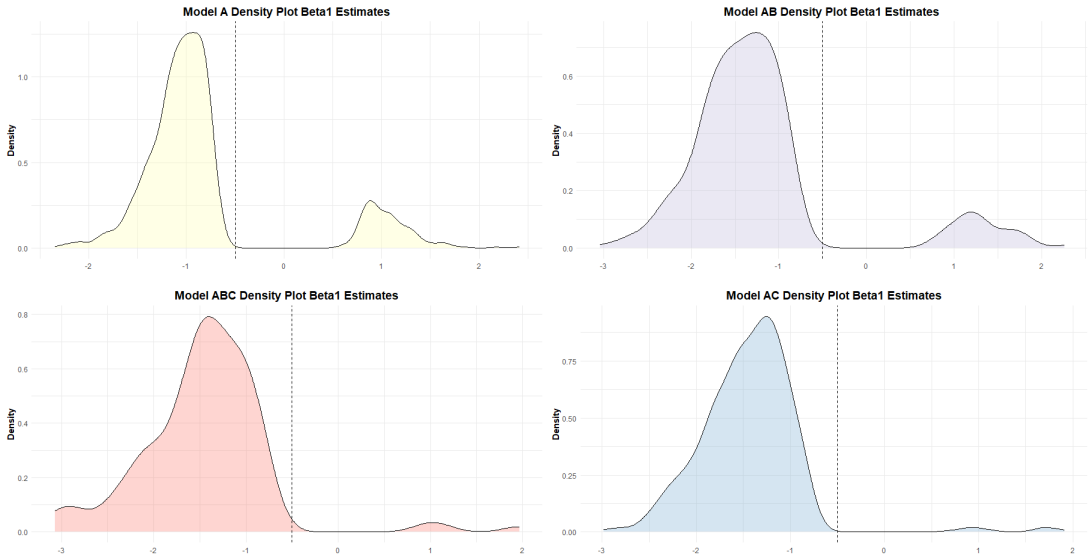

Figure 2: Parameter Setting 1 - $\hat{\beta}_1$ Coefficient for each model

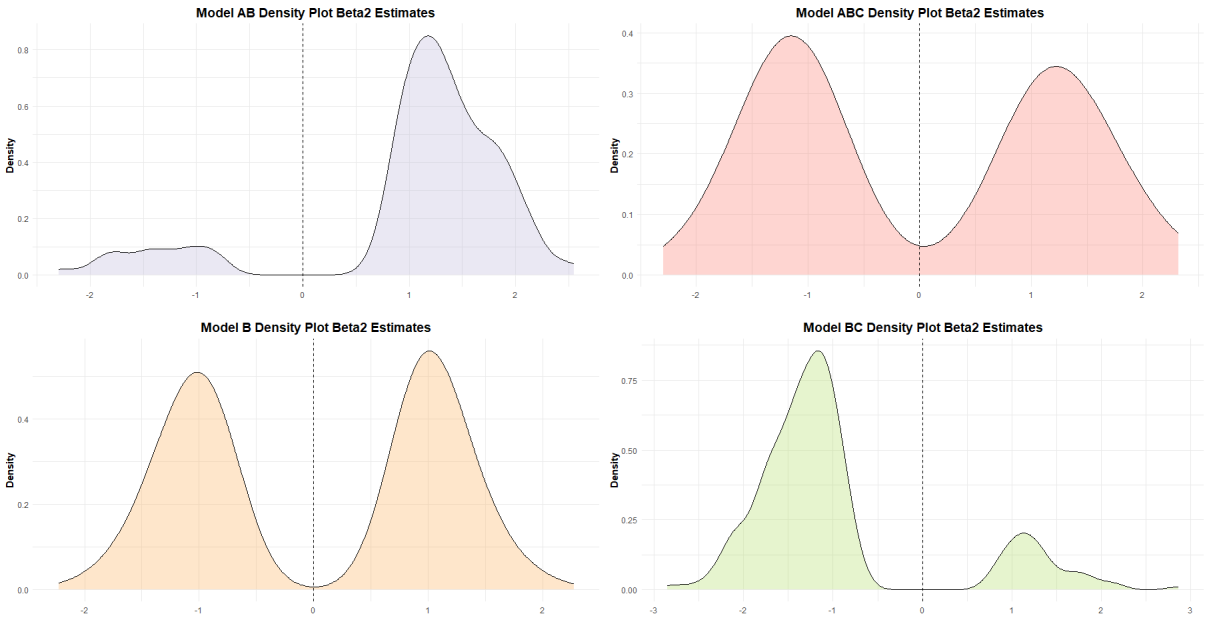

Figure 3: Parameter Setting 1 - $\hat{\beta}_2$ Coefficient for each model

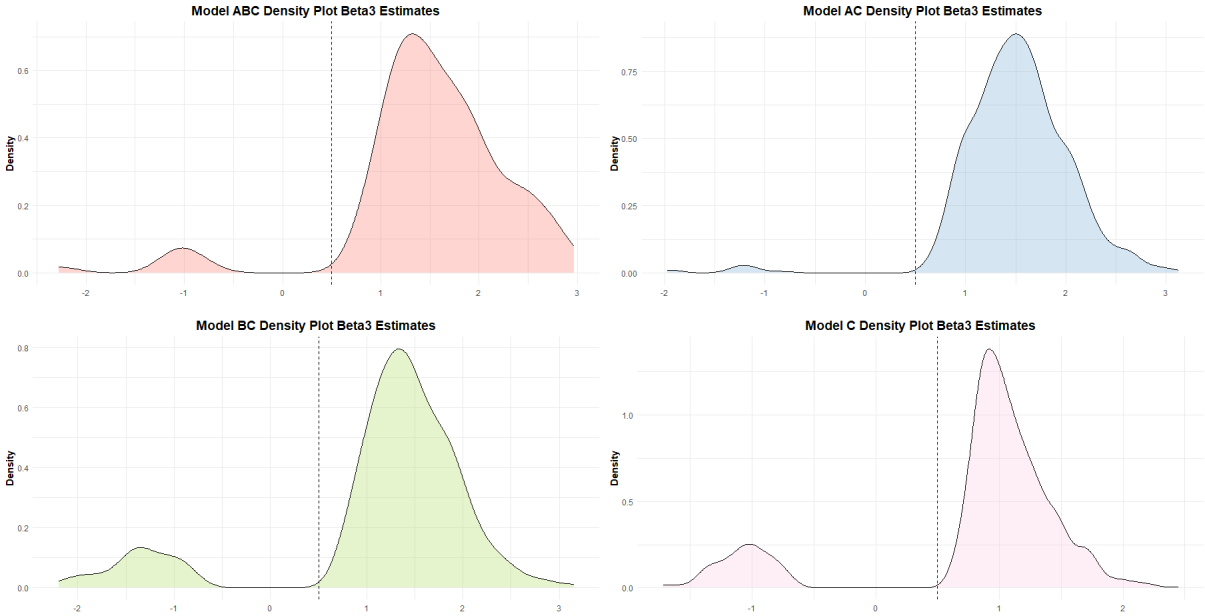

Figure 4: Parameter Setting 1 - $\hat{\beta}_3$ Coefficient for each model

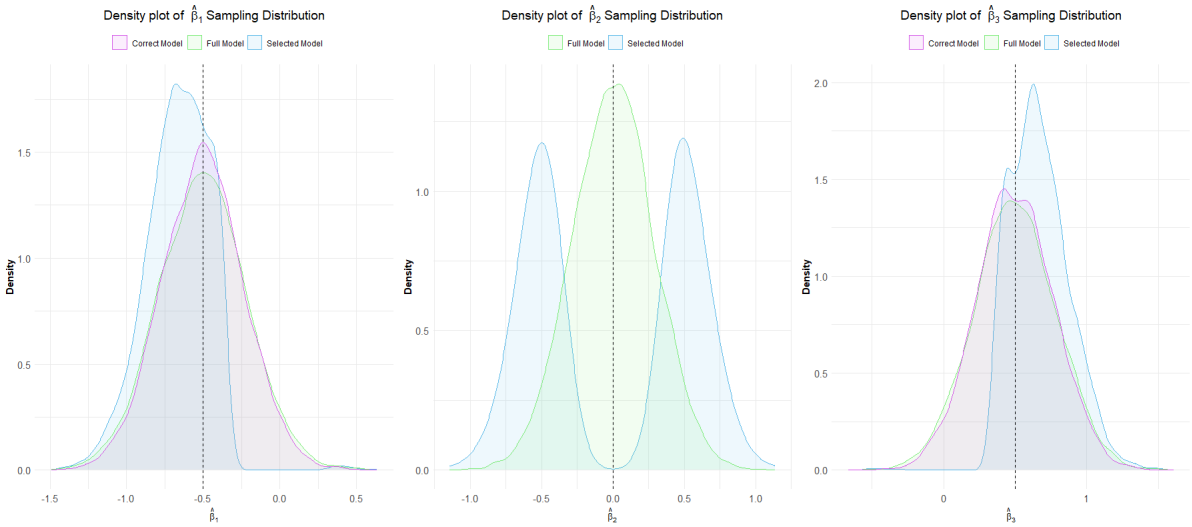

Figure 5: Parameter Setting 1.2 - Mixture of each coefficient across chosen models

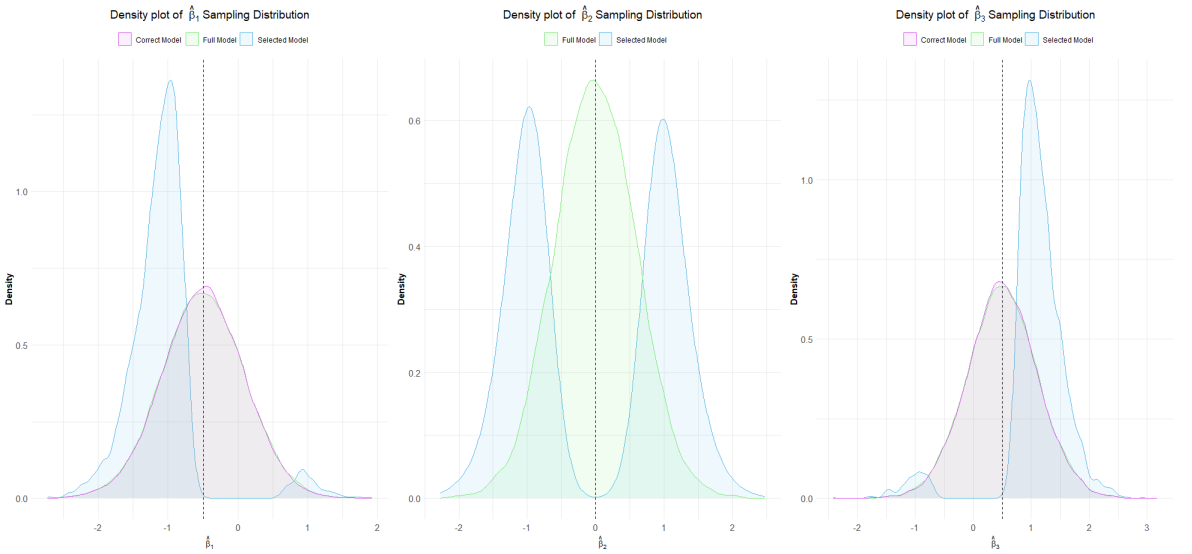

Figure 6: Parameter Setting 1.3 - Mixture of each coefficient across chosen models

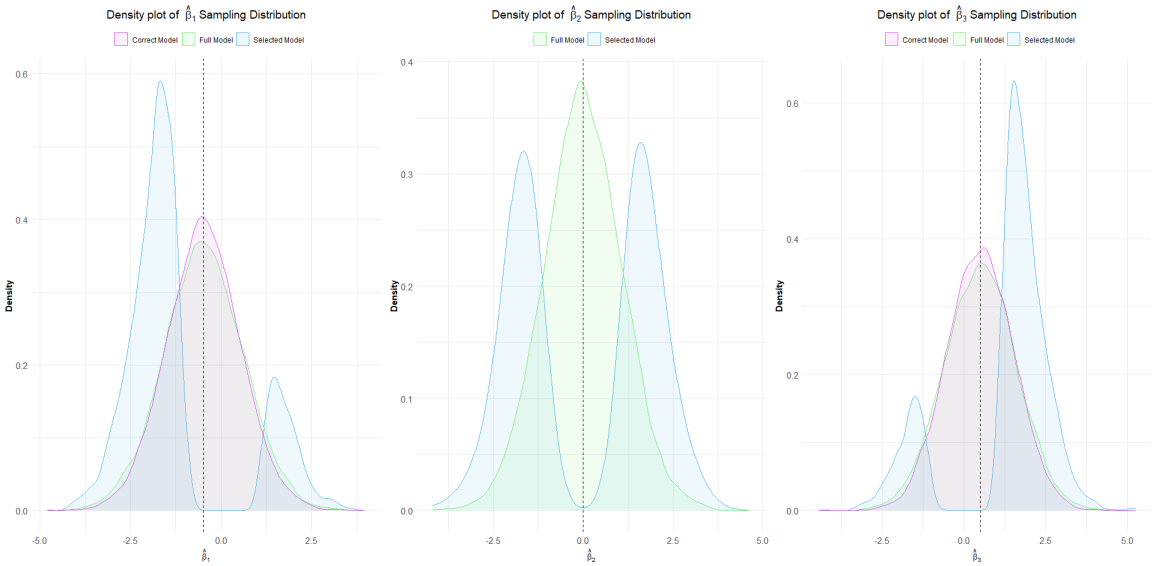

Figure 7: Parameter Setting 1.4 - Mixture of each coefficient across chosen models

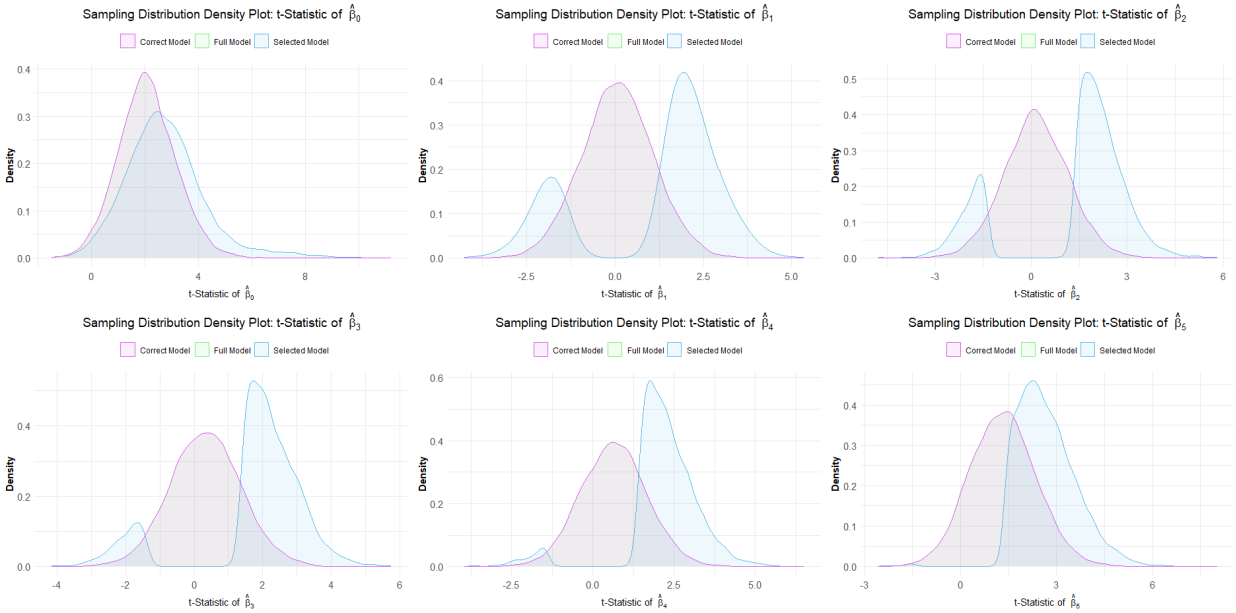

Figure 8: Parameter Setting 2.1 - Mixture of each coefficient across chosen models

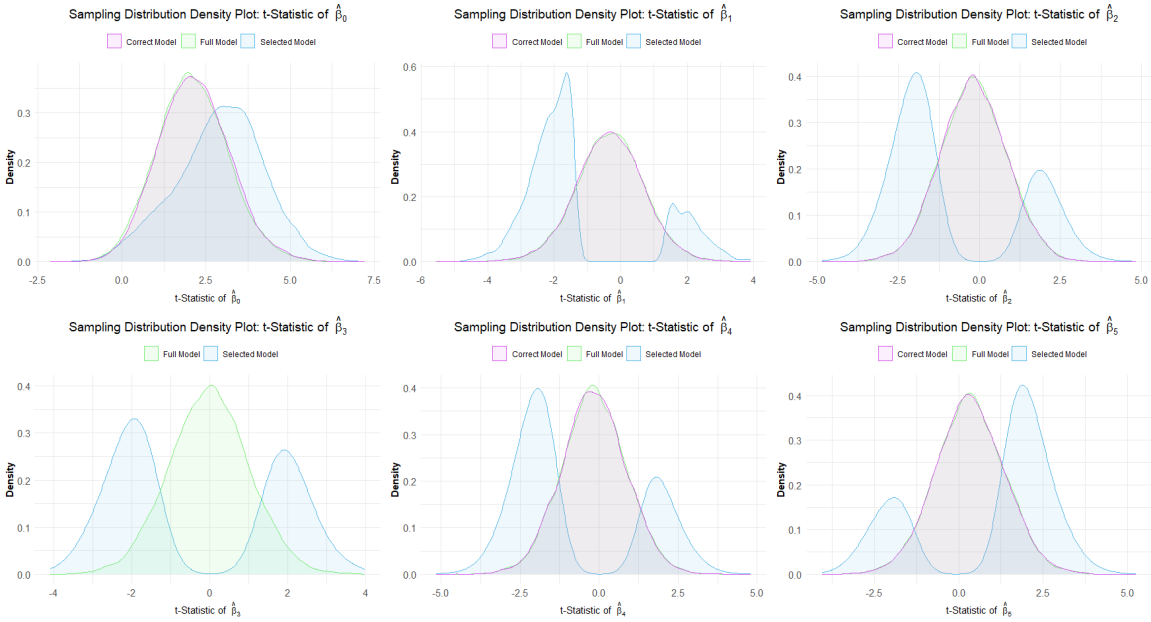

Figure 9: Parameter Setting 2.2 - Mixture of each coefficient across chosen models

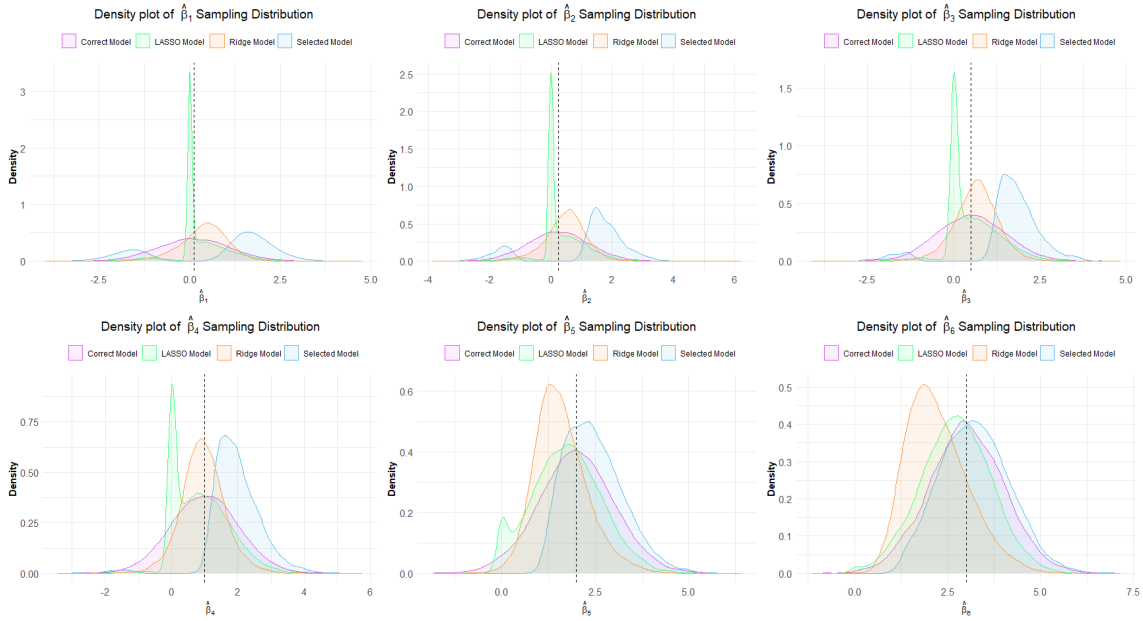

Figure 10: Parameter Setting 3 - Mixture of each coefficient across chosen models

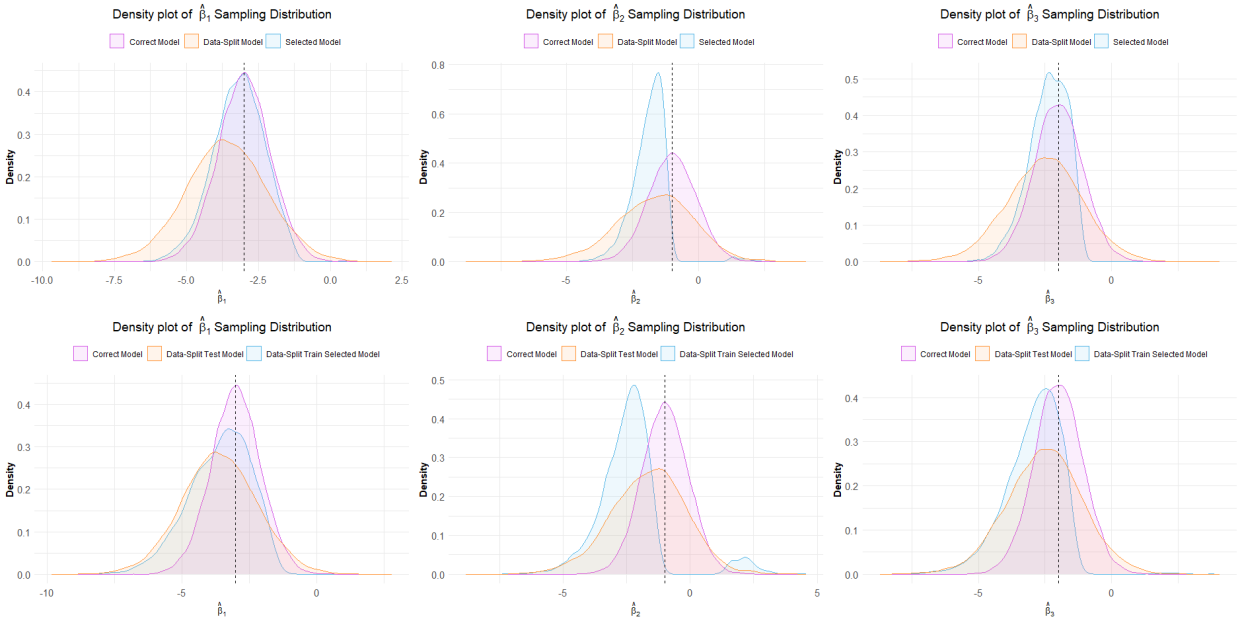

Figure 11: Parameter Setting 4 - Mixture of each coefficient across chosen models (Data-splitting 50-50)

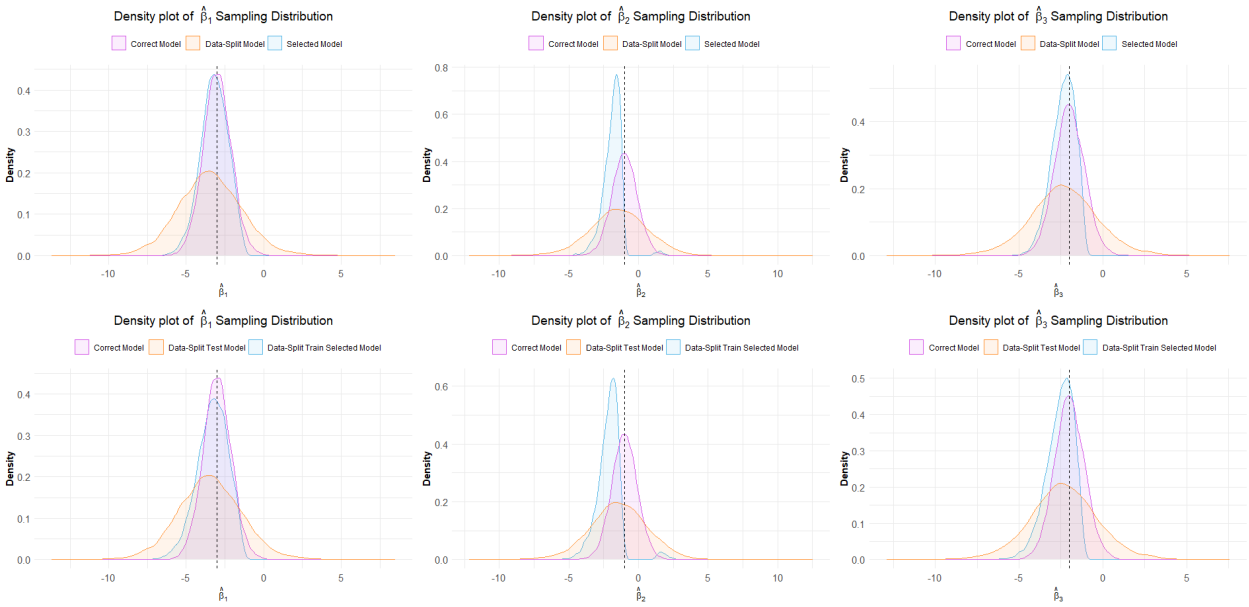

Figure 12: Parameter Setting 4 - Mixture of each coefficient across chosen models (Data-splitting 75-25)

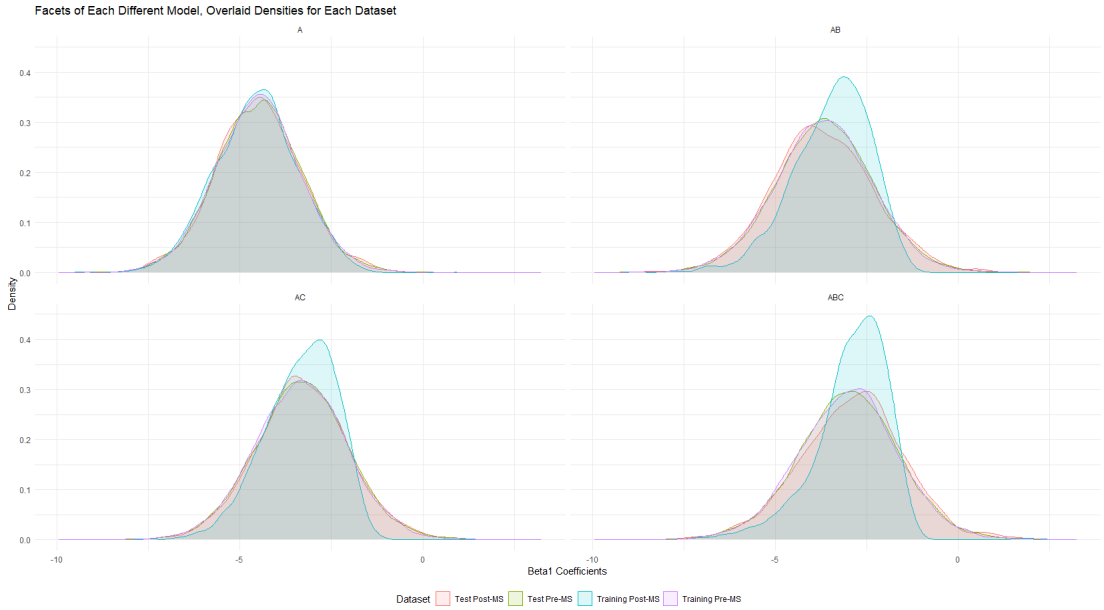

Figure 13: Parameter Setting 4 - Decomposition of 50-50 Data-Split - Estimated Coefficients for β_1

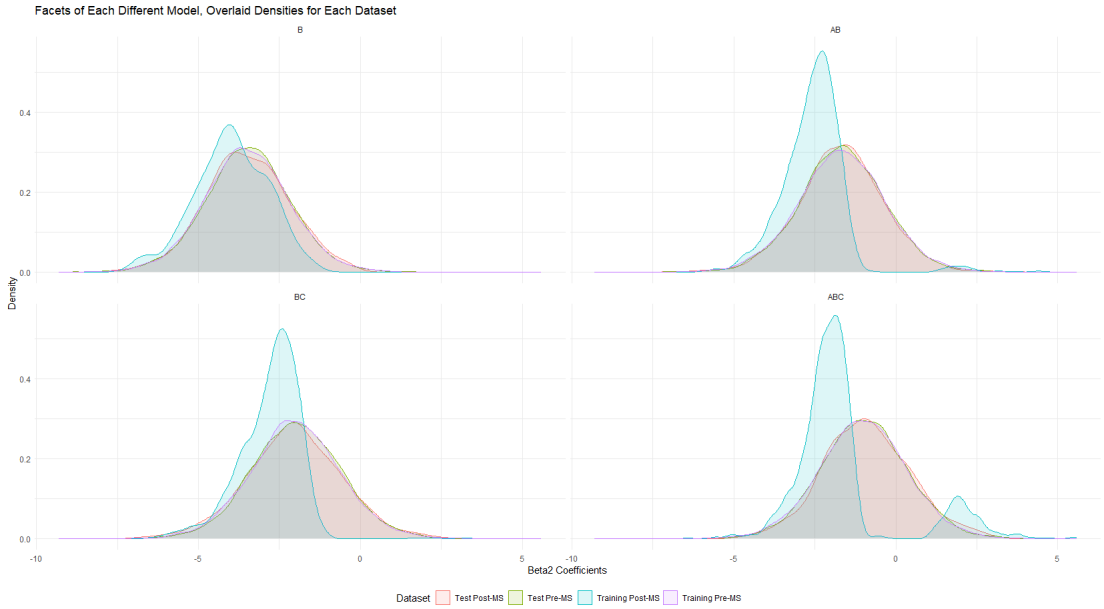

Figure 14: Parameter Setting 4 - Decomposition of 50-50 Data-Split - Estimated Coefficients for β_2

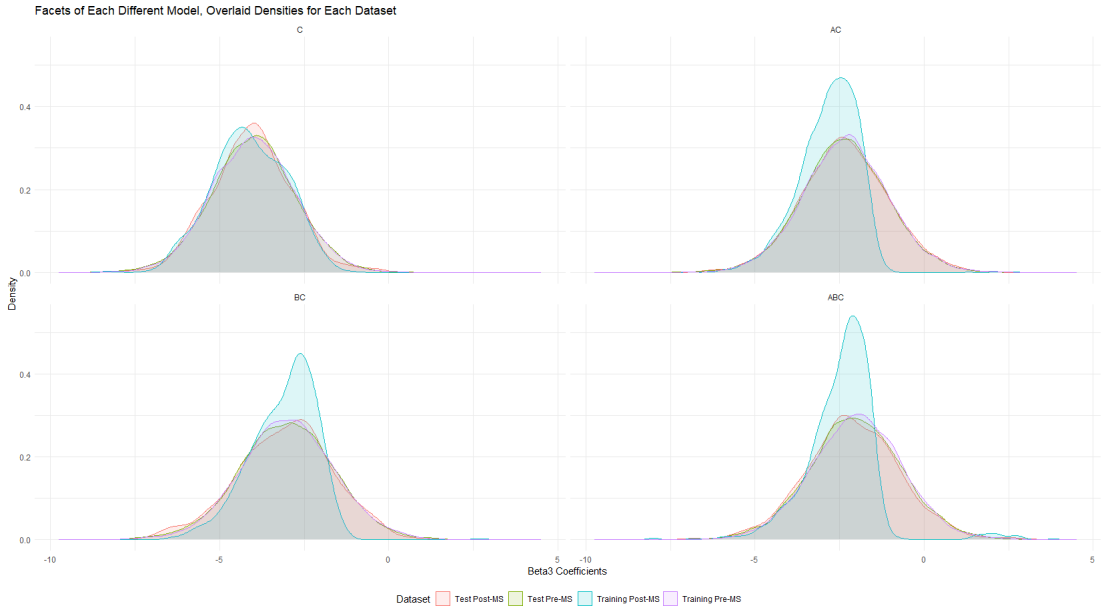

Figure 15: Parameter Setting 4 - Decomposition of 50-50 Data-Split - Estimated Coefficients for β_3

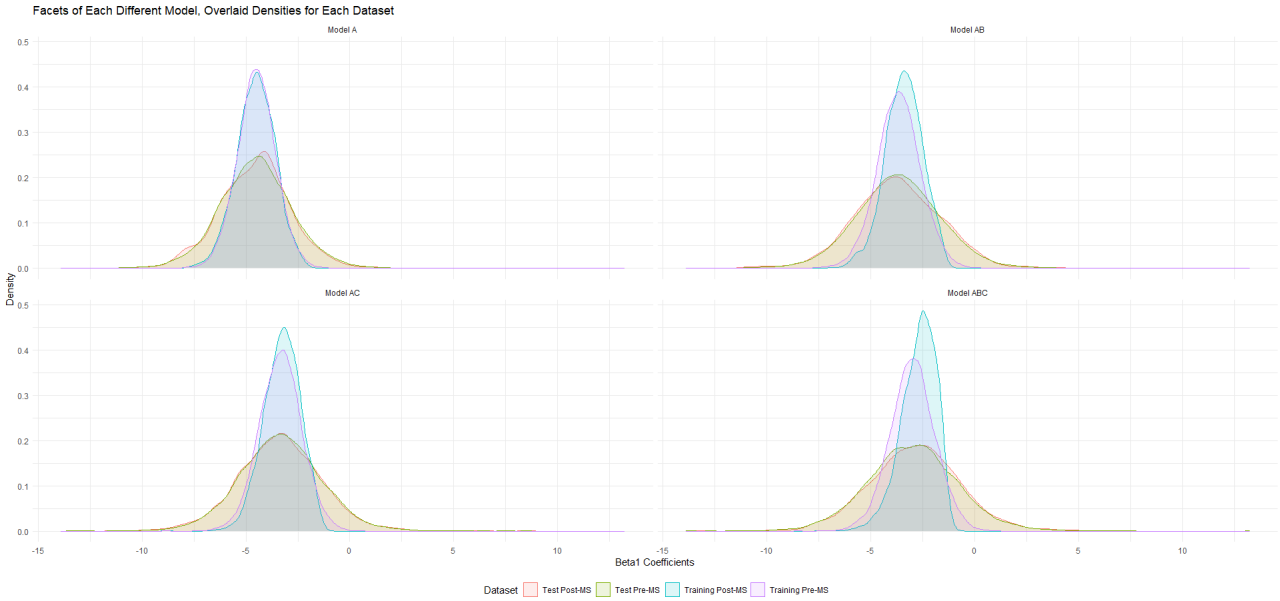

Figure 16: Parameter Setting 4 - Decomposition of 75-25 Data-Split - Estimated Coefficients for β_1

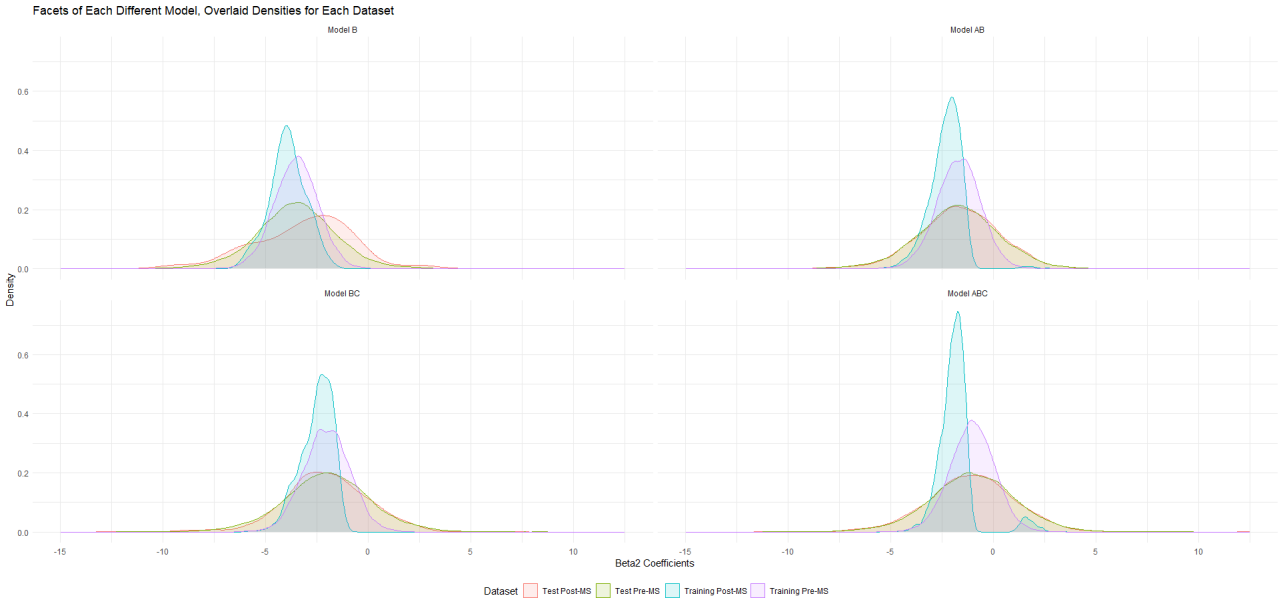

Figure 17: Parameter Setting 4 - Decomposition of 75-25 Data-Split - Estimated Coefficients for β_2

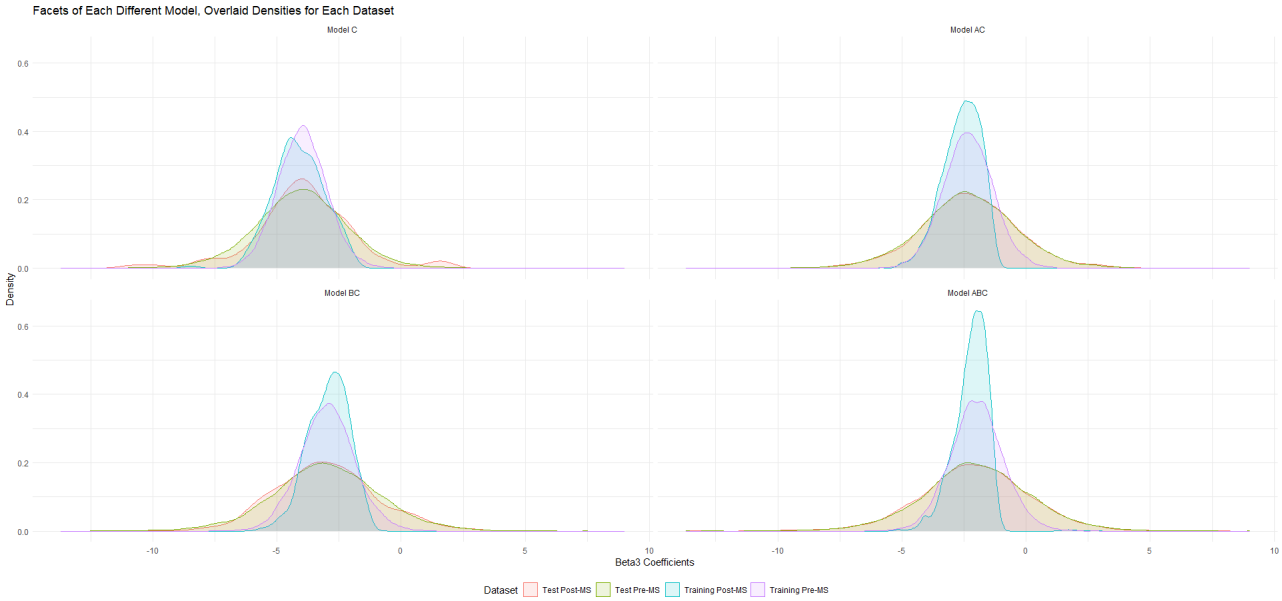

Figure 18: Parameter Setting 4 - Decomposition of 75-25 Data-Split - Estimated Coefficients for β_3